



Modelling Financial Markets

Seminar 3: Factor Models and Asset Pricing — Hands-on



Daniel Traian PELE
Antoaneta Amza

Bucharest University of Economic Studies
IDA Institute Digital Assets
Blockchain Research Center
AI4EFin Artificial Intelligence for Energy Finance
Romanian Academy, Institute for Economic Forecasting
MSCA Digital Finance

Acronyms used in this material (1/2)

- ▣ AI: Artificial Intelligence
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BH: Benjamini–Hochberg (procedure)
- ▣ BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- ▣ CAPM: Capital Asset Pricing Model
- ▣ CMA: Conservative Minus Aggressive (investment factor)
- ▣ CML: Capital Market Line
- ▣ ETF: Exchange-Traded Fund
- ▣ FDR: False Discovery Rate
- ▣ FF3: Fama–French three-factor model
- ▣ FF5: Fama–French five-factor model
- ▣ FWER: Family-Wise Error Rate
- ▣ GLS: Generalised Least Squares
- ▣ GMM: Generalized Method of Moments
- ▣ GRS: Gibbons–Ross–Shanken (test)
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ HML: High Minus Low (value factor)
- ▣ KRS: Kan–Robotti–Shanken (standard errors)
- ▣ MOM: MOMentum factor
- ▣ NW: Newey–West (standard errors)
- ▣ OLS: Ordinary Least Squares
- ▣ PC1: first Principal Component
- ▣ PC2: second Principal Component
- ▣ PC3: third Principal Component
- ▣ PCA: Principal Component Analysis
- ▣ RF: Risk-Free rate

Acronyms used in this material (2/2)

- **RMW: Robust Minus Weak (profitability factor)**
- **SMB: Small Minus Big (size factor)**
- **US: United States**

Outline

Introduction

Key Definitions

Part A: Derivations

Part B: Estimation and Inference

Part C: Open Question

Wrap-up

Seminar Objectives

By the end of this seminar you can

- ▣ Derive the CAPM, the Sharpe-ratio form of GRS and the Vasicek posterior mean
- ▣ Estimate betas and alphas with robust and bootstrap inference
- ▣ Test the CAPM jointly (GRS) and estimate risk premia (Fama–MacBeth) with misspecification-robust errors
- ▣ Compare factor models and choose the number of latent factors

- ▣ Background: Lecture 3; textbook: [Campbell, Lo & MacKinlay \(1997\)](#), ch. 5–6
- ▣ Seminar: PhD candidate Antoaneta Amza, antoaneta.amza@csie.ase.ro

Structure

- ▣ **Part A:** eight derivations (A1–A8)
- ▣ **Part B:** eleven computational exercises with inference, US and Romanian data (B1–B11)
- ▣ **Part C:** one open question, a starting point for the team project (C1), and an AI answer to critique (C2)

How to Work in This Seminar

- **[Solved]** problems show the full worked solution; **[Proposed]** problems show only the task: their solutions are discussed in class
- Every Part B exercise ends with an **interpretation question**
 - ▶ It cannot be answered by reading a number: ask what would change the conclusion
- Report every estimate with its uncertainty (standard error, interval or p -value)
- Before any regression: align the calendars and look for data errors (unadjusted corporate events)

Solved and Proposed Problems

Solved: worked examples in this deck

- Part A: A1 (CAPM from the tangency portfolio), A3 (Vasicek as a posterior mean), A5 (Bonferroni, Holm, BH, step by step), A7 (errors in variables in pass 2)
- Part B: B1 (beta and alpha of XLK with HAC and bootstrap), B2 (GRS on sector ETFs), B3 (PCA and the number of factors)

Proposed: you solve them

- Part A: A2 (model: A1); A4 (model: A3); A6 (model: A7); A8 (model: B3)
- Part B: B4, B6, B10 (model: B1); B5, B9 (model: B2 and Lecture 3, Section 5); B7 (model: A5); B8 (model: B3); B11 (model: B2)
- Part C: C1, open question (model: B1, B5); C2, critique an AI answer (model: C1)

Setup: Python Environment and Data

-  **Open in Colab:** [chapter3_seminar_notebook.ipynb](#)
- Google Colab (recommended) or local Python ≥ 3.10 with `numpy`, `pandas`, `scipy`, `matplotlib`

```
import numpy as np, pandas as pd
URL = "https://raw.githubusercontent.com/danpele/MFM/main/data/market/"
x1k = pd.read_csv(URL + "XLK.US.csv", index_col="date",
                  parse_dates=True)["adjusted_close"]
FF = "https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/ftp/"
# F-F_Research_Data_5_Factors_2x3_CSV.zip
```

- ETFs and stocks: `adjusted_close`; BET: `close`; factors: [Kenneth French Data Library](#) (percent, divide by 100)
- Joint analyses: join the *prices* on common days first, then take returns

Key Definitions: CAPM and Beta

- ▣ **Excess return** $R_{i,t}^e = R_{i,t} - R_{f,t}$; **CAPM** (Sharpe (1964)): $\mathbb{E}[R_i^e] = \beta_i \mathbb{E}[R_m^e]$
- ▣ **Beta** $\beta_i = \text{Cov}(R_i, R_m) / \text{Var}(R_m)$; **Jensen's alpha**: intercept of $R_{i,t}^e = \alpha_i + \beta_i R_{m,t}^e + \varepsilon_{i,t}$
- ▣ **Portfolio beta** $\beta_p = \sum_i w_i \beta_i$ (betas are linear in the weights)
- ▣ **Blume** (Blume (1971)): $\beta^B = 0.33 + 0.67 \hat{\beta}$
- ▣ **Vasicek** (Vasicek (1973)): $\beta^V = w \hat{\beta} + (1 - w) \bar{\beta}$, $w = \sigma_{\beta}^2 / (\sigma_{\beta}^2 + \text{se}(\hat{\beta})^2)$
- ▣ **HAC** (Heteroskedasticity and Autocorrelation Consistent) standard errors, Newey & West (1987): Bartlett weights $1 - l / (L + 1)$

Key Definitions: Tests and Latent Factors

- ▣ **GRS (Gibbons, Ross & Shanken (1989))**: $W = \frac{T-N-1}{N} \frac{\hat{\alpha}^\top \hat{\Sigma}^{-1} \hat{\alpha}}{1 + \hat{\mu}_m^2 / \hat{\sigma}_m^2} \sim F(N, T - N - 1)$ under H_0 : all $\alpha_i = 0$, exact for i.i.d. Normal errors independent of the factors (Lecture 3)
- ▣ **Fama–MacBeth (Fama & MacBeth (1973))**: monthly cross-sectional slopes $\hat{\lambda}_t$, averaged; **Shanken (Shanken (1992))** scales the non-factor part of their variance by $(1 + c)$, $c = \hat{\lambda}^\top \Sigma_f^{-1} \hat{\lambda}$:
 $\text{Var}_{\text{Sh}} = (1 + c)(\text{Var}_{FM} - \Sigma_f / T) + \Sigma_f / T$
- ▣ **Kan–Robotti–Shanken (Kan, Robotti & Shanken (2013))**: both passes as one GMM system; errors robust to a misspecified model
- ▣ **Bonferroni**: $p_j \leq \alpha / M$; **Holm (Holm (1979))**: $p_{(j)} \leq \alpha / (M - j + 1)$ step by step; **BH (Benjamini & Hochberg (1995))**: largest j with $p_{(j)} \leq j\alpha / M$
- ▣ **PCA (Principal Component Analysis)**: eigenvalues λ_k and eigenvectors of the correlation matrix; share λ_k / N ; number of factors: **Bai & Ng (2002)**, **Ahn & Horenstein (2013)**
- ▣ **Moving-block bootstrap (Künsch (1989))**: resample blocks of consecutive observations, keeping the dependence inside each block

Exercise A1: The CAPM from the Tangency Portfolio [Solved]

Task

- ▣ **Setting:** N risky assets with expected excess returns $\boldsymbol{\mu}$ and positive definite covariance matrix $\boldsymbol{\Sigma}$, plus a risk-free asset
- ▣ **Assumption:** the tangency portfolio $\mathbf{w}_m = \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}/(\mathbf{1}^\top \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu})$ is the market portfolio, $R_m = \mathbf{w}_m^\top \mathbf{R}$
- ▣ (a) Express the covariance vector $\text{Cov}(\mathbf{R}, R_m) = \boldsymbol{\Sigma}\mathbf{w}_m$ as a function of $\boldsymbol{\mu}$
- ▣ (b) Compute $\mu_m = \mathbf{w}_m^\top \boldsymbol{\mu}$ and $\sigma_m^2 = \mathbf{w}_m^\top \boldsymbol{\Sigma} \mathbf{w}_m$ and find the relation between them
- ▣ (c) Combine (a) and (b) to prove $\mu_i = \beta_i \mu_m$ for every asset, with $\beta_i = \text{Cov}(R_i, R_m)/\sigma_m^2$
- ▣ (d) Explain why the same result holds for any efficient portfolio on the capital market line (except the risk-free asset itself), and what changes on the frontier of risky assets only
- ▣ **Report:** the derivation in three steps and one sentence for (d)

Exercise A1: Solution [Solved]

Solution

- ▣ Step 1: let $c = 1^\top \Sigma^{-1} \mu$, a scalar. Then $\Sigma w_m = \Sigma \Sigma^{-1} \mu / c = \mu / c$, so $\text{Cov}(R, R_m) = \mu / c$
- ▣ Step 2: $\sigma_m^2 = w_m^\top \Sigma w_m = w_m^\top \mu / c = \mu_m / c$, hence $c = \mu_m / \sigma_m^2$
- ▣ Step 3: $\mu = c \text{Cov}(R, R_m) = \mu_m \text{Cov}(R, R_m) / \sigma_m^2 = \beta \mu_m$: the security market line
- ▣ (d) A CML portfolio holds aw_m , $a > 0$, in risky assets: its excess return is aR_m^e , so $\beta_{i,p} = \beta_i / a$, $\mu_p = a\mu_m$ and $\mu_i = \beta_{i,p}\mu_p$ again
- ▣ Risky-asset frontier portfolios have $w \propto \Sigma^{-1}(\mu - \gamma 1)$: the same steps give $\mu_i - \gamma = \beta_{i,p}(\mu_p - \gamma)$, with the zero-beta rate $R_f + \gamma$ of [Black \(1972\)](#) instead of R_f . Either way a CAPM test also tests the efficiency of the market proxy (Roll's critique)

Exercise A2: GRS Equals a Sharpe-Ratio Gain [Proposed]

Solution discussed in class

Task

- ▣ **Setting:** one factor f (the market excess return) and N test assets R ; maximum-likelihood moments $\mu_f, \sigma_f^2, \mu_R, \Sigma_R, \sigma_{Rf}$
- ▣ **Definitions:** $\beta = \sigma_{Rf}/\sigma_f^2$, $\alpha = \mu_R - \beta\mu_f$, residual covariance $\Sigma = \Sigma_R - \beta\beta^\top \sigma_f^2$
- ▣ (a) Write $V = \text{Cov}((f, R^\top)^\top)$ in partitioned form and invert it with the Schur complement of σ_f^2
- ▣ (b) Show that $SR^2(f, R) = \mu_Y^\top V^{-1} \mu_Y$ equals $\mu_f^2/\sigma_f^2 + \alpha^\top \Sigma^{-1} \alpha$, so that $\alpha^\top \Sigma^{-1} \alpha = SR^2(f, R) - SR^2(f)$
- ▣ (c) Write the GRS statistic W as a function of the two Sharpe ratios
- ▣ (d) Compute W for the 25 size $\times$ B/M portfolios: $T = 757$, $N = 25$, monthly $SR(f) = 0.134$, $SR(f, R) = 0.405$; the 5% critical value of $F(25, 731)$ is 1.52
- ▣ **Report:** the derivation, W , the decision at 5% and one sentence on the economic meaning. **Model:** A1

Exercise A3: Vasicek as a Posterior Mean [Solved]

Task

- ▣ **Data:** 9 sector ETFs (XLB, XLE, XLF, XLI, XLK, XLP, XLU, XLV, XLY), monthly excess returns on SPY, January 1999 – December 2012; OLS betas with Newey–West errors
- ▣ **Numbers:** XLK $\hat{\beta} = 1.415$, $\text{se} = 0.139$; cross-section: mean $\bar{\beta} = 0.949$, variance of the estimated betas 0.116, average squared standard error 0.0077; realised XLK beta in 2013–2026: 1.172
- ▣ **Assumptions:** prior $\beta_i \sim N(\bar{\beta}, \sigma_{\beta}^2)$; likelihood $\hat{\beta}_i | \beta_i \sim N(\beta_i, \text{se}_i^2)$ with se_i known
- ▣ (a) Derive the posterior distribution of β_i given $\hat{\beta}_i$ and its mean β_i^V
- ▣ (b) Use the law of total variance to show $\text{Var}_{cs}(\hat{\beta}) = \sigma_{\beta}^2 + \mathbb{E}[\text{se}^2]$, and write the empirical-Bayes estimator of σ_{β}^2
- ▣ (c) Compute σ_{β}^2 , the weight w and β^V for XLK, and the Blume beta $0.33 + 0.67\hat{\beta}$
- ▣ (d) Compare the absolute errors of the raw, Vasicek and Blume betas against the realised 1.172
- ▣ **Report:** the formulas, the four numbers of (c) and the three errors of (d)

Exercise A3: Solution (1/2) [Solved]

Solution (1/2)

- (a) Posterior density $\propto \exp\left\{-\frac{(\beta - \bar{\beta})^2}{2\sigma_{\beta}^2}\right\} \exp\left\{-\frac{(\hat{\beta} - \beta)^2}{2\text{se}^2}\right\}$: the exponent is quadratic in β , so the posterior is Normal
- Its precision is $1/\sigma_{\beta}^2 + 1/\text{se}^2$ and its mean is $\frac{\bar{\beta}/\sigma_{\beta}^2 + \hat{\beta}/\text{se}^2}{1/\sigma_{\beta}^2 + 1/\text{se}^2} = w\hat{\beta} + (1 - w)\bar{\beta}$, with $w = \frac{\sigma_{\beta}^2}{\sigma_{\beta}^2 + \text{se}^2}$
- (b) $\text{Var}(\hat{\beta}) = \text{Var}(\mathbb{E}[\hat{\beta} \mid \beta]) + \mathbb{E}[\text{Var}(\hat{\beta} \mid \beta)] = \text{Var}(\beta) + \mathbb{E}[\text{se}^2]$
- Hence $\hat{\sigma}_{\beta}^2 = \text{Var}_{cs}(\hat{\beta}) - \overline{\text{se}^2}$; using $\text{Var}_{cs}(\hat{\beta})$ itself counts the sampling noise as true dispersion and shrinks too little

Exercise A3: Solution (2/2) [Solved]

Solution (2/2)

- ▣ (c) $\sigma_{\beta}^2 = 0.116 - 0.008 = 0.108$ and $se^2 = 0.139^2 = 0.0193$
- ▣ $w = 0.108 / (0.108 + 0.0193) = 0.849$, so $\beta^V = 0.849 \times 1.415 + 0.151 \times 0.949 = 1.344$
- ▣ With the raw variance 0.116 the weight would be 0.857 and $\beta^V = 1.348$; Blume: $0.33 + 0.67 \times 1.415 = 1.278$
- ▣ (d) Errors against 1.172: raw 0.243, Vasicek 0.172, Blume 0.106
- ▣ Interpretation: XLK's beta is estimated precisely, so Vasicek shrinks it only a little; Blume moves every beta one third of the way towards 1 ($\beta^B - 1 = 0.67(\hat{\beta} - 1)$) and is closer here. Exercise A4 shows why regression to the mean exceeds sampling noise

Exercise A4: Blume's Slope as a Shrinkage Weight [Proposed]

Solution discussed in class

Task

- ▣ **Data:** betas of the same 9 sector ETFs in two non-overlapping samples, 1999–2012 ($\hat{\beta}_{i,1}$) and 2013–2026 ($\hat{\beta}_{i,2}$); fitted cross-sectional line $\hat{\beta}_2 = 0.230 + 0.724 \hat{\beta}_1$; from A3: $\sigma_\beta^2 = 0.108$, $\overline{se^2} = 0.0077$, $\bar{\beta} = 0.949$
- ▣ **Assumptions:** true betas are constant, $\hat{\beta}_{i,k} = \beta_i + \eta_{i,k}$, the noises are independent of β_i and of each other, with common variance se^2
- ▣ (a) Derive the probability limit, as $N \rightarrow \infty$, of the OLS slope b of $\hat{\beta}_2$ on $\hat{\beta}_1$ and of the intercept
- ▣ (b) Show that b equals the Vasicek weight with a common standard error, so that Blume's rule is a Vasicek estimator
- ▣ (c) Compute the implied slope and intercept from the A3 numbers and compare them with 0.724 and 0.230
- ▣ (d) The fitted slope has standard error 0.114: is the gap significant, and which explanations remain possible?
- ▣ **Report:** the probability limits, the implied numbers and a one-sentence interpretation. **Model:** A3

Exercise A5: Ten Candidate Factors [Solved]

Task

- ▣ **Setting:** ten new candidate factors, each tested separately with a two-sided t -test of zero mean; level $\alpha = 5\%$, $M = 10$
- ▣ **Sorted p -values:** 0.001, 0.004, 0.009, 0.012, 0.021, 0.035, 0.048, 0.060, 0.20, 0.55
- ▣ (a) Count the discoveries of the naive rule $p \leq \alpha$
- ▣ (b) Apply Bonferroni: reject H_j if $p_j \leq \alpha/M$
- ▣ (c) Apply Holm step by step: compare $p_{(j)}$ with $\alpha/(M - j + 1)$ and stop at the first failure
- ▣ (d) Apply Benjamini–Hochberg: find the largest j with $p_{(j)} \leq j\alpha/M$ and reject $H_{(1)}, \dots, H_{(j)}$
- ▣ (e) State which error rate each procedure controls, and under which dependence between the tests
- ▣ **Report:** the four counts and one sentence on which procedure suits a factor search

Exercise A5: Solution [Solved]

Solution

- (a) Seven p -values are at most 0.05: **7** discoveries
- (b) $\alpha/M = 0.005$: only 0.001 and 0.004 pass: **2**
- (c) Holm thresholds 0.005, 0.0056, 0.00625: $0.001 \leq 0.005$, $0.004 \leq 0.0056$, $0.009 > 0.00625$, stop: **2**
- (d) BH thresholds $0.005j$: 0.005, 0.010, 0.015, 0.020, 0.025, 0.030, $\dots$; $p_{(5)} = 0.021 \leq 0.025$, while $p_{(6)} = 0.035 > 0.030$ and no later $p_{(j)}$ passes: **5**
- (e) Bonferroni and Holm control the FWER under any dependence; BH controls the FDR for independent tests or under positive regression dependence on the true nulls; Benjamini–Yekutieli ([Benjamini & Yekutieli \(2001\)](#)), thresholds divided by $\sum_{j=1}^M 1/j = 2.93$ under any dependence: **1** discovery here
- Interpretation: in a factor search the share of false factors (FDR) is the relevant target; among FWER methods Holm always rejects at least as much as Bonferroni

Exercise A6: A Useless Factor in Pass 2 [Proposed]

Solution discussed in class

Task

- ▣ **Setting:** N fixed, $T \rightarrow \infty$; return vectors R_t i.i.d. over time, means μ_i , nonsingular covariance Σ , finite fourth moments; a factor g_t , i.i.d., mean 0, variance σ_g^2 , independent of all returns, so $\beta_{i,g} = 0$
- ▣ (a) Show that the OLS slope $\hat{\beta}_{i,g} = \widehat{\text{Cov}}(R_i, g) / \hat{\sigma}_g^2$ satisfies $\sqrt{T} \hat{\beta}_{i,g} \rightarrow N(0, \sigma_i^2 / \sigma_g^2)$, so $\hat{\beta}_{i,g} = O_p(T^{-1/2})$
- ▣ (b) With g alone in pass 2, write $\hat{\lambda}_g = \widehat{\text{Cov}}_{cs}(\bar{R}_i, \hat{\beta}_{i,g}) / \widehat{\text{Var}}_{cs}(\hat{\beta}_{i,g})$ in terms of $z = \sqrt{T} \hat{\beta}_g$ and show that $\hat{\lambda}_g = O_p(\sqrt{T})$ when the μ_i differ across assets
- ▣ (c) Derive the joint limit of $(\sqrt{T}(\bar{R} - \mu), z)$ and show that when all μ_i are equal, $\hat{\lambda}_g$ converges in distribution to a non-degenerate random variable, not 0
- ▣ (d) Explain why the Fama–MacBeth t -test of $\lambda_g = 0$ over-rejects (Kan & Zhang (1999)) and which first-pass test protects against it
- ▣ **Report:** the orders of magnitude, the recommended check and a comparison with the Lecture 3 simulation. **Model:** A7



Exercise A7: Errors in Variables in Pass 2 [Solved]

Task

- ▣ **Setting:** $R_{it} = a_i + \beta_i f_t + \varepsilon_{it}$ with pricing restriction $a_i = \lambda_0 + \beta_i(\lambda - \mu_f)$, $\mu_f = \mathbb{E}[f_t]$; ε_{it} i.i.d. over t and i , variance σ_ε^2 , independent of f and of the β_i ; β_i i.i.d. across assets with variance $\text{Var}_{cs}(\beta)$
- ▣ Work conditionally on the realised factor path $f_1, \dots, f_T$; pass 2: OLS of $\bar{R}_i$ on $(1, \hat{\beta}_i)$, $\hat{\beta}_i = \beta_i + \eta_i$, $\eta_i = \sum_t (f_t - \bar{f}) \varepsilon_{it} / \sum_t (f_t - \bar{f})^2$
- ▣ (a) Show that $\text{Var}(\eta_i | f) = \sigma_\varepsilon^2 / (T \hat{\sigma}_f^2)$
- ▣ (b) Show that η_i is uncorrelated with $\bar{\varepsilon}_i$ and that, for $N \rightarrow \infty$ and T fixed, the pass-2 slope converges to $\lambda_T \text{Var}_{cs}(\beta) / [\text{Var}_{cs}(\beta) + \sigma_\varepsilon^2 / (T \hat{\sigma}_f^2)]$, with $\lambda_T = \lambda + \bar{f} - \mu_f$ the realised premium
- ▣ (c) Estimate this attenuation factor for the CAPM on the 25 size $\times$ B/M portfolios (monthly 1963–2026, $T = 757$) and on the 9 sector ETFs (1999–2026, $T = 331$), with $\text{Var}_{cs}(\beta) = \text{Var}_{cs}(\hat{\beta}) - \bar{\sigma}_\varepsilon^2 / (T \hat{\sigma}_f^2)$
- ▣ (d) Explain what the Shanken correction fixes and what it does not
- ▣ **Report:** the derivation, the two attenuation factors and one sentence on single stocks

Exercise A7: Solution (1/2) [Solved]

Solution (1/2)

- (a) Given f , η_i is a linear combination of the ε_{it} with weights $(f_t - \bar{f}) / \sum_s (f_s - \bar{f})^2$, so
 $\text{Var}(\eta_i | f) = \sigma_\varepsilon^2 / \sum_t (f_t - \bar{f})^2 = \sigma_\varepsilon^2 / (T \hat{\sigma}_f^2)$
- (b) $\text{Cov}(\bar{\varepsilon}_i, \eta_i | f) = \sigma_\varepsilon^2 \frac{1}{T} \sum_t (f_t - \bar{f}) / \sum_s (f_s - \bar{f})^2 = 0$, because $\sum_t (f_t - \bar{f}) = 0$
- With the restriction, $\bar{R}_i = \lambda_0 + \lambda_T \beta_i + \bar{\varepsilon}_i$: the realised factor mean $\bar{f}$ is common to all assets and does not average out as $N \rightarrow \infty$
- Given f , the pairs $(\beta_i, \eta_i, \bar{\varepsilon}_i)$ are i.i.d. across i , so by the law of large numbers $\widehat{\text{Cov}}_{cs}(\bar{R}, \hat{\beta}) \rightarrow \lambda_T \text{Var}_{cs}(\beta)$ and $\widehat{\text{Var}}_{cs}(\hat{\beta}) \rightarrow \text{Var}_{cs}(\beta) + \sigma_\varepsilon^2 / (T \hat{\sigma}_f^2)$
- The ratio is the stated limit: attenuation of the realised premium λ_T towards zero; $\lambda_T - \lambda = \bar{f} - \mu_f$ vanishes only as $T \rightarrow \infty$

Exercise A7: Solution (2/2) [Solved]

Solution (2/2)

- (c) 25 portfolios: $\text{Var}_{cs}(\hat{\beta}) = 0.01905$, noise $\bar{\sigma}_\varepsilon^2 / (T \hat{\sigma}_f^2) = 0.00062$, $\text{Var}_{cs}(\beta) = 0.01843$; factor $0.01843 / 0.01905 = 0.967$
- 9 sectors: $\text{Var}_{cs}(\hat{\beta}) = 0.09247$, noise 0.00200 , $\text{Var}_{cs}(\beta) = 0.09048$; factor $0.09048 / 0.09247 = 0.978$
- Interpretation: the bias is only 2–3% here, because diversified portfolios have small residual variance and T is long; single stocks have much larger σ_ε^2 , so the attenuation grows
- (d) [Shanken \(1992\)](#) corrects the standard error of $\hat{\lambda}$ for the noise in $\hat{\beta}$ (a variance correction); it does not remove the finite- T bias of the point estimate

Exercise A8: Equicorrelated Assets [Proposed]

Solution discussed in class

Task

- ▣ **Setting:** N standardised returns with equal pairwise correlation $\rho = 0.5$; correlation matrix $C = (1 - \rho)I + \rho\mathbf{1}\mathbf{1}^\top$
- ▣ (a) For $N = 3$, find the eigenvalues and eigenvectors of C
- ▣ (b) Compute the share of total variance explained by the first component, λ_1/N
- ▣ (c) Generalise to N assets and find the limit of the PC1 share as $N \rightarrow \infty$
- ▣ (d) Invert the formula of (c) to obtain the equicorrelation-equivalent correlation of a PC1 share s ; apply it to the BVB blue chips ($N = 10$, $s = 0.425$, B8) and to the US sectors ($N = 11$, $s = 0.660$, B3)
- ▣ (e) For a general correlation matrix with average off-diagonal correlation $\bar{\rho}$, show that $1 + (N - 1)\bar{\rho} \leq \lambda_1$
- ▣ **Report:** eigenvalues, the PC1 share, the limit, the two correlations and the bound. **Model:** B3 and the PCA recap of Lecture 3 (two assets: $\lambda = 1 \pm \rho$)

Exercise B1: Beta and Alpha of XLK [Solved]

Task

- **Data:** XLK (technology ETF) adjusted prices, file `XLK.US.csv` in `data/market`, month-end values; Mkt-RF and RF from the [Kenneth French Data Library](#); monthly excess returns January 1999 – July 2026 ($T = 331$)
- (a) Estimate $R_{\text{XLK},t}^e = \alpha + \beta \text{Mkt-RF}_t + \varepsilon_t$ by OLS and annualise α (multiply by 12)
- (b) Compute classical and Newey–West standard errors of α and β , with Bartlett weights and $L = \lfloor 4(T/100)^{2/9} \rfloor$ lags
- (c) Build a 95% moving-block bootstrap interval for α : blocks of 12 months, 2000 draws, percentile method ([Künsch \(1989\)](#))
- (d) Test $H_0 : \alpha = 0$ at 5% with the Newey–West t -statistic
- **Report:** a table with estimates, classical and Newey–West errors, $t(\alpha)$, R^2 and the bootstrap interval

Interpretation

- The HAC error of beta is twice the classical one: which feature of the residuals causes it?

Exercise B1: Solution (1/2) [Solved]

Solution (1/2)

- (b) Lags: $L = \lfloor 4 \times 3.31^{2/9} \rfloor = \lfloor 5.2 \rfloor = 5$

	Estimate	Classical s.e.	Newey–West s.e.
α (annualised)	0.33%	2.27%	2.45%
β	1.277	0.041	0.082

- (a) $R^2 = 0.742$; the robust 95% interval of β is $1.277 \pm 1.96 \times 0.082 = [1.12, 1.44]$ instead of $[1.20, 1.36]$
- (d) $t(\alpha) = 0.33/2.45 = 0.13$, $p = 0.89 > 0.05$: do not reject; no evidence of a CAPM alpha
- (c) Bootstrap: $\alpha \in [-5.5\%, 4.9\%]$, standard error 2.61%; the interval contains 0, as the Newey–West test says

Exercise B1: Solution (2/2) [Solved]

Solution (2/2)

- Interpretation: classical errors assume $\text{Var}(\varepsilon_t \mid f_t)$ is constant
- The sandwich estimator weights each squared residual by $(f_t - \bar{f})^2$: residual variance is high exactly when the market moves a lot (tech bust 2000–2002, 2008, 2020)
- This heteroskedasticity makes the classical formula understate $\text{Var}(\hat{\beta})$: the ratio of errors is $0.082/0.041 = 2.0$
- The point estimates are the same; only the uncertainty changes

Exercise B2: GRS Test on Sector ETFs [Solved]

Task

- ▣ **Data:** 9 sector ETFs (XLB, XLE, XLF, XLI, XLK, XLP, XLU, XLV, XLY), month-end adjusted prices joined on common months; monthly excess returns January 1999 – July 2026 ($T = 331$); market factor Mkt-RF
- ▣ (a) Estimate the 9 time-series regressions on Mkt-RF and report the annualised alphas and the betas
- ▣ (b) Compute the GRS statistic with the maximum-likelihood residual covariance and its p -value from $F(9, 321)$
- ▣ (c) Compute the monthly Sharpe ratio of the market and of the tangency portfolio of market plus sectors, and check the identity of A2
- ▣ (d) Decide at the 5% level
- ▣ **Report:** alphas, betas, GRS, p -value and the two Sharpe ratios

Interpretation

- ▣ The same test rejects the CAPM on the 25 size $\times$ B/M portfolios (GRS = 4.20). Why is it silent here?

Exercise B2: Solution (1/2) [Solved]

Solution (1/2)

- ▣ (a) Alphas: XLU +3.4%, XLE +2.4%, XLV +1.8%, XLP +1.7%, XLF -2.3% a year; the other four lie within $\pm 0.6\%$
- ▣ Betas: XLU 0.40, XLP 0.45, XLV 0.67, XLE 0.92, XLI 1.04, XLB 1.05, XLF 1.07, XLY 1.07, XLK 1.28
- ▣ (b) $GRS = 0.53$, $p = 0.85$
- ▣ (c) Monthly Sharpe ratios: market 0.146, market plus sectors 0.191 (annual 0.51 and 0.66)
- ▣ Identity: $W = \frac{321}{9} \times \frac{0.191^2 - 0.146^2}{1 + 0.146^2} = 35.67 \times 0.0150 = 0.53$
- ▣ (d) $p = 0.85 > 0.05$: do not reject H_0 : all $\alpha_i = 0$

Exercise B2: Solution (2/2) [Solved]

Solution (2/2)

- Interpretation: the test has little power on these assets
- Sectors mix small and big, value and growth stocks, so they do not spread the characteristics whose premia the CAPM misses; their alphas are small and similar
- The Sharpe-ratio gain from the sectors is small ($0.51 \rightarrow 0.66$ a year), against $0.47 \rightarrow 1.40$ for the 25 portfolios
- Not rejecting is not accepting the CAPM: choose test assets that spread the characteristics suspected to be priced

Exercise B3: PCA and the Number of Factors [Solved]

Task

- ▣ **Data:** 11 sector ETFs (the 9 above plus XLRE and XLC), daily log returns on common days, 20 June 2018 – 18 September 2026 ($T = 2073$); SPY for comparison
- ▣ (a) Standardise the returns; compute the correlation matrix, its eigenvalues and eigenvectors (sign chosen so that the average loading is positive)
- ▣ (b) Report the variance shares of PC1–PC3 and the loadings of PC1 and PC2
- ▣ (c) Correlate the PC1 scores with the SPY returns
- ▣ (d) Choose the number of factors with Kaiser ($\lambda_k > 1$), Bai & Ng (2002) (IC_{p2}) and the eigenvalue ratio of Ahn & Horenstein (2013), with $k_{\max} = 5$
- ▣ **Report:** shares, loadings, the correlation and the three estimates of k

Interpretation

- ▣ PC2 opposes utilities and staples to technology: would it earn a premium? What would you need to test it?

Exercise B3: Solution (1/2) [Solved]

Solution (1/2)

- (a) Eigenvalues: 7.26, 1.06, 0.82, 0.40, 0.32, ...
- (b) Shares: PC1 66.0%, PC2 9.7%, PC3 7.4%; the first three explain 83.1%
- PC1 loadings are all positive (0.24–0.34); PC2: XLU +0.55, XLP +0.42, XLRE +0.31 against XLK −0.40, XLY −0.33: defensive versus growth
- (c) $\text{corr}(\text{PC1}, \text{SPY}) = 0.954$: the first latent factor is the market

Exercise B3: Solution (2/2) [Solved]

Solution (2/2)

- (d) Kaiser: 2 (only 7.26 and 1.06 exceed 1)
- Bai–Ng: IC_{p2} is minimised at $k = 5 = k_{\max}$; the penalty $\frac{N+T}{NT} \ln N$ is small when $N = 11$
- Ahn–Horenstein: ratios 6.82, 1.30, 2.05, 1.26, 1.15, maximum at $k = 1$
- The estimators disagree: with $N = 11$ the large- N asymptotics do not apply, so report the range, not a single k
- Interpretation: PCA explains covariance, not average returns. To test a premium, run Fama–MacBeth with the PC2 loadings as betas on many assets, or sort assets on PC2 loadings into a long–short portfolio and test its mean with Newey–West errors

Exercise B4: CAPM Betas of BVB Blue Chips [Proposed]

Solution discussed in class

Task

- ▣ **Data:** 11 BVB blue chips (TLV, SNP, BRD, TGN, FP, SNG, H2O, SNN, EL, DIGI, TEL: <SYMBOL>.R0.csv) and BET.csv; each stock aligned separately with BET; daily log returns, 6 January 2015 (or listing) – 18 September 2026; constant R_f cancels
- ▣ (a) For each pair, remove days with $|r_t| > 0.5$ for the stock or BET (unadjusted corporate events); report how many
- ▣ (b) Estimate each beta by OLS on BET, with Newey–West errors (lag rule) and R^2
- ▣ (c) Compute empirical-Bayes Vasicek betas (A3), the 11 stocks as cross-section
- ▣ (d) Compute Dimson betas (Dimson (1979)): regress $r_{i,t}$ on $r_{\text{BET},t-1}$, $r_{\text{BET},t}$, $r_{\text{BET},t+1}$ and sum the slopes (leads and lags formed before removing days)
- ▣ **Report:** table of $\hat{\beta}$, s.e., R^2 , Vasicek and Dimson betas; chart with 95% intervals. **Model:** B1, A3

Interpretation

- ▣ How does asynchronous trading of less liquid stocks bias daily betas, and what does the Dimson column show?

Exercise B5: Fama–MacBeth on ETFs [Proposed]

Solution discussed in class

Task

- ▣ **Data:** 16 test assets, the 9 sector ETFs and 7 factor ETFs (MTUM, VVALUE, QUAL, USMV, IWM, IWD, IWF); monthly excess returns August 2013 – July 2026 ($T = 156$); factors Mkt-RF, SMB, HML
- ▣ (a) Pass 1: estimate full-sample time-series betas of each ETF on the three factors
- ▣ (b) Pass 2: run 156 monthly cross-sectional OLS regressions of returns on $(1, \hat{\beta}_i)$ and average the slopes
- ▣ (c) Compute Fama–MacBeth, Shanken, $\text{Var}_{\text{sh}} = (1 + c)(\text{Var}_{FM} - \Sigma_f / T) + \Sigma_f / T$, and Kan–Robotti–Shanken errors (Kan, Robotti & Shanken (2013))
- ▣ (d) Compare each $\hat{\lambda}$ with the time-series mean of its factor and test $\lambda = 0$ at 5%
- ▣ **Report:** a table with $\hat{\lambda}$, the three standard errors and the factor means. **Model:** B2 and Lecture 3, Section 5

Interpretation

- ▣ Is a non-significant premium here evidence that size and value are not priced?

Exercise B6: Do Factor ETFs Have Alpha? [Proposed]

Solution discussed in class

Task

- **Data:** adjusted prices of MTUM and QUAL (data/market); daily FF5 (Fama & French (2015)) and MOM (Carhart (1997)) factors; each ETF on its own trading calendar
- (a) Build daily excess returns, keeping only returns between consecutive factor trading days
- (b) Estimate the six-factor regression and annualise α (multiply by 252)
- (c) Compute the Newey–West standard error (10 lags) and the t -statistic of α ; test $\alpha = 0$ at 5%
- (d) Build a 95% moving-block bootstrap interval for α : blocks of 20 days, 1000 draws
- **Report:** n , α , Newey–West error, t and the bootstrap interval for each ETF. **Model:** B1

Interpretation

- Why must a 3-day ETF return never be regressed on a 1-day factor return?

Exercise B7: Multiple Testing of 25 Alphas [Proposed]

Solution discussed in class

Task

- **Data:** the 25 size $\times$ B/M portfolios and the FF3 factors (Fama & French (1993)), monthly excess returns July 1963 – July 2026 ($T = 757$)
- (a) Estimate the 25 FF3 time-series regressions; compute the Newey–West t -statistic of each alpha (lag rule) and the two-sided p -value from the Normal distribution
- (b) Count the discoveries at 5%: naive, Bonferroni, Holm and Benjamini–Hochberg, and the number with $|t| > 3$ (Harvey, Liu & Zhu (2016))
- (c) Identify the portfolio with the largest $|t|$ and report its alpha
- **Report:** the five counts and the alpha of the portfolio with the largest $|t|$, together with its t -statistic. **Model:** A5

Interpretation

- The 25 tests are strongly correlated: which methods remain valid, and which test uses the correlation directly?

Exercise B8: PCA of BVB Blue Chips [Proposed]

Solution discussed in class

Task

- ▣ **Data:** 10 blue chips (H2O excluded: listed in 2023) and BET; daily log returns on common days from 7 June 2017; remove days with any $|r_t| > 0.5$
- ▣ (a) Standardise the returns; compute the eigenvalues, the variance shares and the PC1 loadings
- ▣ (b) Correlate the PC1 scores with the BET returns
- ▣ (c) Choose the number of factors with Kaiser, Bai–Ng (IC_{p2}) and Ahn–Horenstein, with $k_{\max} = 5$
- ▣ (d) Compare the PC1 share and the number of factors with the US sectors (B3)
- ▣ **Report:** shares, loadings, the correlation and the three estimates of k . **Model:** B3

Interpretation

- ▣ What does a weaker first factor imply for diversification within the BVB?

Exercise B9: A Lewellen–Nagel–Shanken Check (1/2) [Proposed]

Solution discussed in class

Task (1/2)

- **Data:** 25 size $\times$ B/M portfolios and 30 industry portfolios (Kenneth French Data Library), monthly excess returns July 1963 – July 2026 ($T = 757$); models CAPM, FF3 and FF5 (Lewellen, Nagel & Shanken (2010))
- (a) Estimate pass-1 betas; regress the mean returns on $(1, \hat{\beta}_i)$ by OLS and compute the OLS R^2
- (b) Repeat by GLS with weight $\hat{V}^{-1}$ (sample covariance of the returns): $R_{GLS}^2 = 1 - \mathbf{e}^\top \hat{V}^{-1} \mathbf{e} / (\boldsymbol{\mu} - \bar{\boldsymbol{\mu}}_{GLS})^\top \hat{V}^{-1} (\boldsymbol{\mu} - \bar{\boldsymbol{\mu}}_{GLS})$

Exercise B9: A Lewellen–Nagel–Shanken Check (2/2) [Proposed]

Solution discussed in class

Task (2/2)

- (c) Build 95% intervals by an i.i.d. bootstrap over months (500 draws), re-estimating both passes in each draw
- (d) Repeat (a)–(c) on the 55 assets; for FF3 also impose $\lambda = \mathbb{E}[f]$ with no intercept
- **Report:** a table of OLS and GLS R^2 with intervals for the 3 models and 2 sets of assets. **Model:** B2 and Lecture 3, Section 5

Interpretation

- Does the FF3 R^2 survive once industries are added?

 MFM_ch3_two_pass_inference

Exercise B10: Comparing Models by Sharpe Ratios (1/2) [Proposed]

Solution discussed in class

Task (1/2)

- **Data:** monthly factors Mkt-RF, SMB, HML, RMW, CMA, MOM ([Kenneth French Data Library](#)), July 1963 – July 2026 and January 2000 – July 2026; models CAPM, FF3, Carhart (FF3 + MOM), FF5 and FF5 + MOM ([Barillas & Shanken \(2018\)](#)); FF3 and Carhart use the three-factor SMB, FF5 the five-factor SMB
- (a) Compute each model's maximum squared Sharpe ratio $SR^2 = \mu_f^\top \Sigma_f^{-1} \mu_f$, annualise it (multiply by 12) and report its square root
- (b) Build 95% percentile intervals for the SR^2 differences FF3 – CAPM, FF5 + MOM – FF5 (nested), FF5 – FF3 and FF5 – Carhart (non-nested), with a stationary bootstrap ([Politis & Romano \(1994\)](#)), mean block 6 months, 2000 draws

Exercise B10: Comparing Models by Sharpe Ratios (2/2) [Proposed]

Solution discussed in class

Task (2/2)

- (c) Regress each of the six factors on the other five (spanning regressions) and report the Newey–West t of the intercept
- (d) Decide which factors show no significant incremental contribution at 5% in each period
- **Report:** the Sharpe-ratio table, the four intervals and the six spanning t per period. **Model:** B1

Interpretation

- Why is a bootstrap interval useless for nested models? Which factors add no significant alpha after 2000?

 MFM_ch3_two_pass_inference

Exercise B11: GRS without Normality (1/2) [Proposed]

Solution discussed in class

Task (1/2)

- ▣ **Data:** 25 size $\times$ B/M portfolios, CAPM with Mkt-RF, monthly July 1963 – July 2026 ($T = 757$, $N = 25$)
- ▣ (a) Estimate the 25 regressions; compute the kurtosis of each residual series and the average kurtosis
- ▣ (b) Compute GRS with the maximum-likelihood $\hat{\Sigma}$ and its p -value from $F(25, 731)$

Exercise B11: GRS without Normality (2/2) [Proposed]

Solution discussed in class

Task (2/2)

- (c) Residual bootstrap under H_0 : resample whole rows of residuals (keeping their cross-correlation), rebuild $R_t^* = \hat{\beta}f_t + \varepsilon_t^*$, recompute GRS 2000 times; report the p -value and the 95% bootstrap critical value; valid for non-Normal errors that are i.i.d. and independent of the factors
- (d) GMM Wald test on the moments $\varepsilon_t \otimes (1, f_t)$: $W = \hat{\alpha}^\top \hat{V}_\alpha^{-1} \hat{\alpha} \rightarrow \chi_{25}^2$, $\hat{V}_\alpha$: estimate of $\text{Var}(\hat{\alpha})$, White and Newey–West (6 lags)
- (e) Explain why GRS cannot be computed for the 100 size $\times$ B/M portfolios over August 2018 – July 2026 ($T = 96$)
- **Report:** the kurtosis, the four p -values and the critical values. **Model:** B2

Interpretation

- Is the rejection of the CAPM an artefact of assuming Normal residuals?

C1: Do Factor Premia Exist on the Bucharest Stock Exchange? [Proposed]

Solution discussed in class

Task

- **Question:** do momentum (Jegadeesh & Titman (1993)) and low beta (Frazzini & Pedersen (2014)) earn premia among BVB blue chips?
- **Data:** month-end adjusted prices of the 10 blue chips of B8 and BET, December 2015 – August 2026 (September 2026 is incomplete); monthly simple returns; a month with $|r| > 0.5$ is a data error: return 0 in the signal and in the holding month (rule fixed in advance)
- (a) Build the 12–1 momentum signal (return from month $t - 12$ to $t - 2$); a stock is eligible in month t if its signal and its price at the end of $t - 1$ exist
- (b) Each month, go long the 3 stocks with the highest signal and short the 3 with the lowest, with equal weights
- (c) Low beta: estimate 36-month rolling betas on BET (at least 18 months of data); go long the 3 lowest-beta and short the 3 highest-beta stocks
- (d) Report the mean, the volatility and the Newey–West t (3 lags) of each long–short return; compare t with 1.96 and with the $|t| > 3$ hurdle of the factor zoo
- (e) Discuss the number of stocks, survivorship (today's blue chips chosen with hindsight), short-selling constraints, costs and liquidity; name public sources of book equity and shares outstanding
- **Deliverable:** a Quantlet folder in the team repository and one slide with the conclusion and its main limitation

C2: Critique an AI Answer (1/2) [Proposed]

Solution discussed in class

- **The prompt** a student sent to an AI (Artificial Intelligence) assistant: “P holds cleaned month-end adjusted prices of the BVB blue chips of C1, one column per stock. Build a 12-month momentum long–short portfolio, top 3 minus bottom 3 with equal weights, and report its annual mean and volatility”
- **The answer:** the code below, followed by the conclusion

```
R      = np.log(P).diff()                # monthly returns
mom    = P.pct_change(12)                # past 12-month return
ls     = {}
for t in R.index[13:]:
    s = mom.loc[t].dropna().sort_values() # rank the stocks
    ls[t] = R.loc[t, s.index[-3:]].mean() - R.loc[t, s.index[:3]].mean()
ls = pd.Series(ls)
print(12 * ls.mean(), np.sqrt(12) * ls.std()) # annual mean, volatility
```

- **Conclusion of the AI:** “Momentum earns about 50% a year with a volatility of 17%. This agrees with [Fama & French \(2012\)](#), who document momentum on the Bucharest Stock Exchange.”

C2: Critique an AI Answer (2/2) [Proposed]

Solution discussed in class

Task

- ▣ (a) Find the errors: there are two or three; the rest of the answer is correct
- ▣ (b) For each error, say how you would detect it: a test, a published number, the lecture or the paper itself
- ▣ (c) Give the corrected code and the numbers it produces
- ▣ (d) Log each error as in `AI_ERRORS.md`: what was asked, what the AI answered, what was wrong, how you found out, the correct version
- ▣ Model: C1 and Lecture 3

Why this exercise

- ▣ AI code usually runs without an error message: the mistakes are in the finance, not in the syntax
- ▣ Course rule: you may use AI, declare it in `AI_USE.md` and log at least three errors you caught in `AI_ERRORS.md`

Common Pitfalls

Data

- ▣ Unadjusted corporate events (spin-offs, capital returns) create fake returns
- ▣ Regressing multi-day asset returns on one-day factors
- ▣ Forgetting that French data are in percent
- ▣ Using signals that were not known at the portfolio date

Statistics

- ▣ Classical errors for betas under volatility clustering
- ▣ Fama–MacBeth errors without the Shanken or misspecification correction
- ▣ Trusting a high cross-sectional R^2 on 25 portfolios
- ▣ $|t| > 1.96$ after trying many factors
- ▣ Reading “not rejected” as “true”

- ▣ Rule: state the sample, the frequency, the market proxy and the risk-free rate of every regression

Take-Home Assignment (Team Project)

- Choose **five** assets: at least two BVB stocks, one US sector ETF and one factor ETF
- For each asset
 - ▶ CAPM beta with Newey–West errors, a rolling beta and its Vasicek-adjusted value
 - ▶ For US assets: FF5 + MOM loadings and alpha with a bootstrap interval
- GRS and PCA by market, not across markets: US assets on Mkt-RF (USD, monthly, RF from the French files); BVB stocks on BET (RON, monthly, constant R_f); with 2–3 assets per test, power is low
- Compare with Lecture 3
- Deliverables: a GitHub repository with a notebook that runs top-to-bottom and a 2-page summary
- Textbook for PCA: [Härdle & Simar \(2019\)](#)
- Log every AI error you catch, in this seminar or in the project, in `AI_ERRORS.md`

Further Study on Quantinar

Advanced courses

- ▣  Generalized Method of Moments
- ▣  Multivariate Statistical Analysis

Code

- ▣ **Notebook:** `chapter3_seminar_notebook.ipynb` (Colab)
- ▣ **Quantlets:** github.com/danpele/MFM/tree/main/Quantlets/Ch_03

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